



# Dimitri Lerévérend

*Inria Rennes*

## Education

- 2023-now **Ph.D Student**, *INRIA Rennes*, Rennes, Davide Frey, Romaric Gaudel and François Taïani  
Exploring privacy in decentralized learning using noise correlation, both from a theoretical and a practical point a view.
- 2022-2023 **Master's degree (M2 SIF) in computer science**, *ENS Rennes*, Rennes, France  
Focus on variety of topics: machine learning, image representation, complex systems proofs.
- 2021-2022 **Preparation for the "Agrégation d'informatique"**, *ENS Rennes*, Rennes, France  
Focus on teaching and the core of computer science : Algorithmics, Architecture, Operating Systems, Networking, Databases.
- 2019-2020 **Bachelor's degree in computer science**, *École Normale Supérieure de Rennes*, Rennes, France. Ranking: 7/28

## Professional experience

- 2023-now **Ph.D. Student**, *INRIA Rennes*, Rennes  
Explore noise correlation in differentially private decentralized learning.
- 2023 (5 months) **Research Internship**, *Privacy Protection through private averaging*, Team WIDE, INRIA Rennes, Davide Frey, François Taïani and Romaric Gaudel.  
Studied privacy properties of decentralized machine learning and the effects of private averaging using Pytorch.
- 2021 (3 months) **Research Intership**, *Adapting Public Announcement Logic update mechanisms to Dynamic Epistemic Logic*, D7, IRISA Rennes, Guillaume Aucher
- 2020-2021 **Research project**, *Feature extraction for high dynamic range images*, Percept, INRIA Rennes, Olivier Le Meur and Mathieu Chambe  
Created a neural network for feature extraction of HDR images in Tensorflow.
- 2020 (2 months) **Research Intership**, *Visualisation of echo states network*, Mnemosyne, INRIA Bordeaux, Nicolas Rouger  
Creation of spatialized ESN and visualisation of the activity with Voronoi diagrams.

## Publications

- [1] **Bellet, Aurélien and Cyffers, Edwige and Frey, Davide and Gaudel, Romaric and Lerévérend, Dimitri and Taïani, François**, *Unified privacy guarantees for decentralized learning via matrix factorization*, International Conference on Learning Representations, ICLR 2026

- [2] **Biswas, Sayan and Frey, Davide and Gaudel, Romaric and Kermarrec, Anne-Marie and Lerévérend, Dimitri and Pires, Rafael and Sharma, Rishi and Taïani, François**, Low-Cost Privacy-Preserving Decentralized Learning, Proceedings on Privacy Enhancing Technologies 2025
- [3] **Biswas, Sayan and Frey, Davide and Gaudel, Romaric and Gupta, Nirupam and Kermarrec, Anne-Marie and Lerévérend, Dimitri and Pires, Rafael and Sharma, Rishi and Taïani, François and de Vos, Martijn**, *Mosaic Learning: A Framework for Decentralized Learning with Model Fragmentation*, arXiv:2602.04352

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## Diplomas/Certifications

- 2023 **Master's degree**, *University of Rennes*
- 2023 **Agrégation d'informatique**, *French national education*, Ranked 2nd.

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## Teachings

- 2023-2025 **ACE, L1**, ISTIC  
Course explaining algorithm complexity practically.
- 2023-2024 **BDD, L2**, ISTIC  
Course presenting databases.
- 2024-2026 **PRI, M1**, Cyberschool  
Course presenting privacy formally.
- 2024-2025 **Thrustworthy AI, M1**, Master IA  
Course presenting risks in AI: privacy risks, attacks against AI models.

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## Programming languages

- Main Python, OCaml, Latex, C
- Notions C++, Java, R
- Libraries sklearn, Tensorflow, pytorch, pandas

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## Languages

- 1<sup>st</sup> language French
- fluent English